

Raspberry Pi setup instructions

Common steps for master and clients

Connect the Ethernet cable to the Raspberry.
Power on the Raspberry.
Connect the Raspberry to the internet.

The following commands must be run in the Raspberry Pi terminal :
Ping test → to test the connection

```
ping -c 3 google.com
```

Update the Raspberry system

```
sudo apt update  
sudo apt full-upgrade -y
```

Install the following modules and verify the status (active and enable)

Chrony : `sudo apt install chrony -y` then `sudo systemctl status chrony`
OpenCV : `sudo apt install opencv -y` then `sudo systemctl status opencv`
Rpicam : `sudo apt install rpicam -y` then `sudo systemctl status rpicam`

Open rpicam → adjust the camera focus
`rpicam-hello --timeout 0`

Create a systemd service → automatically launch the master or client program at startup

```
sudo nano /etc/systemd/system/lauch.service
```

MASTER

Once inside the service configuration file, write the following :

[Unit]

```
Description=Launch Python Service  
After=network-online.target  
Wants=network-online.target
```

[Service]

```
ExecStartPre=/bin/sh -c '> /home/raspberry/log.txt'  
ExecStartPre=/bin/sh -c '> /home/raspberry/logError.txt'  
ExecStart=/usr/bin/python3 /home/raspberry/mastercmd3.py  
StandardOutput=file:/home/raspberry/log.txt  
StandardError=file:/home/raspberry/logError.txt  
Restart=on-failure  
User=root
```

[Install]

```
WantedBy=multi-user.target
```

Access the crontab

```
crontab -e
```

→ at the very end of the file, add the following line :

```
@reboot sudo find /media/raspberry -type d -empty -delete
```

CLIENT

Once inside the service configuration file, write the following :

[Unit]

```
Description=Start camera script  
After=network-online.target  
Wants=network-online.target
```

[Service]

```
ExecStartPre=/bin/sh -c '> /home/Kth/log.txt'  
ExecStartPre=/bin/sh -c '> /home/Kth/logError.txt'  
ExecStart=/usr/bin/python3 /home/Kth/slaverecord4.py  
WorkingDirectory=/home/Kth/  
StandardOutput=file:/home/Kth/log.txt  
StandardError=file:/home/Kth/logError.txt  
Restart=no  
User=root
```

[Install]

```
WantedBy=multi-user.target
```

Access the crontab

```
crontab -e
```

→ at the very end of the file, add the following lines

```
@reboot chronyd
```

```
***** /usr/bin/chronyc -a makestep
```

Turn the Wi-Fi off

Have the clock automatically synchronized on clients

On the Raspberry terminal :

```
sudo nano /etc/chrony/chrony.conf
```

Write at the very end of the file :

```
leapsectz right/UTC
```

```
server 'Master IP' iburst minpoll 4 maxpoll 6
```

Write and verify that there's : `^*'Master IP'` :

```
chrony sources -v
```

Then restart chrony :

```
sudo systemctl restart chrony
```

Accessing VNC to view the Raspberry Pi on a laptop

On the Raspberry terminal :

```
sudo systemctl enable wayvnc  
sudo systemctl status wayvnc
```

→ verify that nothing is listed as "disabled"

```
sudo systemctl restart wayvnc  
sudo ss -tulnp | grep 5900
```

→ check that there is a line containing 5900

```
hostname -I
```

→ Note the 2nd IP address

On the laptop :

Open RealVNC Viewer and enter :

```
'2nd IP address noted' : 5900
```

Enter the Raspberry username and password